

Technology #160

apply a marine-inspired anti-biofouling micro-textured surface (e.g., shark-skin riblet pattern) to the catheter lumen in conjunction with the existing antithrombotic coating, creating a physical barrier that hinders protein and platelet adhesion that leads to heparin-binding thrombus

Engineering Technology-Transfer Dossier
CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

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Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

apply a marine■inspired anti■biofouling micro■textured surface (e.g., shark■skin riblet pattern) to the catheter lumen in conjunction with the existing antithrombotic coating, creating a physical barrier that hinders protein and platelet adhesion that leads to heparin■binding thrombus

WHY DOES IT MATTER?

Documented problem 'heparin-binding thrombus formation' concerning tunneled hemodialysis catheter. User-submitted engineering problem (session input): "Why do tunneled hemodialysis catheters develop heparin-binding thrombus despite antithrombotic luminal coatings?". [fda_maude] 100 records retrieved live (1992-2070). Illustrative records: 'MAUDE report 1000: Injury [SAME AS ABOVE]'; 'MAUDE report 1000009: Injury [SLIP-CATH BEACON TIP CATHETER]'. LIMITS STAMPED INLINE: voluntary/regulatory reports; counts are NOT incidence rates; causality UNVERIFIED (Art. XXI.5). [europepmc] 10 records retrieved live (2025-2026). Illustrative records: 'Antithrombotic Polymers: A Narrative Review on Current and Future Strategies for Their Design, Synthesis, and Application.'; 'Polymeric Fibrous Materials for Procoagulant and Anticoagulant Applications: A Review of Molecular Blood-Material Mechanisms and Strategies.'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 11 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in phage_microbio. 7-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 1 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 7-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: perform an in vitro blood flow loop comparison measuring thrombus mass on standard antithrombotic-coated catheters versus catheters with.... Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-003 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: phage infectivity retention vs time on surface. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: kill kinetics on an immobilized surface are not well-mixed growth — re-measure before quantitative use; surface-immobilized phage dosing is not well-mixed — use contact-area density instead (MODEL_DERIVED adaptation); k_{ads} is strain- and environment-specific; surface-immobilized phage dosing is not a well-mixed MOI — use contact-area density instead (MODEL_DERIVED adaptation); BIO-002 (Exponential population dynamics) is applicable only under: unlimited substrate; constant net rate over the window.; BIO-003 (Surface-limited adsorption kinetics) is applicable only under: well-mixed suspension; constant k_{ads} .; BIO-001 (Multiplicity of infection) is applicable only under: well-mixed system; planktonic or defined-surface assay.; Model input variables requiring sourced values before any numeric result: N_0 (-), r (1/s), t (s), k_{ads} (m^3/s), N_{phage} (-), $N_{bacteria}$ (-), N_{phage} (-), $N_{bacteria}$ (-) — all UNKNOWN (no sourced value exists; Art. VIII).; Model output variables produced by the governing model: N (-), $dN_{infected}/dt$ (1/s), MOI (-) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): surface phage loading density [N_{phage} , -] value_status=UNKNOWN; infectivity retention half-life [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; host range breadth [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; moi at contact surface [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; CFU challenge load [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; resistance emergence rate [r , 1/s] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

1 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: biofilm formation

POTENTIAL VALUE DRIVER: Addresses: Documented problem 'heparin-binding thrombus formation' concerning tunneled hemodialysis catheter. User-submitted engineering problem (session input): "Why do tunneled hemodialysis catheters develop heparin-binding thrombus despite antithrombotic luminal coatings?". [fda_maude] 100 records retrieved live (1992-2070). Illustrative records: 'MAUDE report 1000: Injury [SAME AS ABOVE]'; 'MAUDE report 1000009: Injury [SLIP-CATH BEACON TIP CATHETER]'. LIMITS STAMPED INLINE: voluntary/regulatory reports; counts are NOT incidence rates; causality UNVERIFIED (Art. XXI.5). [europepmc] 10 records retrieved live (2025-2026). Illustrative records: 'Antithrombotic Polymers: A Narrative Review on Current and Future Strategies for Their Design, Synthesis, and Application.'; 'Polymeric Fibrous Materials for Procoagulant and Anticoagulant Applications: A Review of Molecular Blood-Material Mechanisms and Strategies.'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: phage_microbio

Disciplines: microbiology, surface engineering, pharmacodynamics, materials science

Architecture: PROPOSED architecture (no physical system exists): phage loading / immobilization layer; release or contact-kill mechanism; surface substrate (coating architecture); stability preservation element — realizing: apply a marine-inspired anti-biofouling micro-textured surface (e.g., shark-skin riblet pattern) to the catheter lumen in conjunction with the existing antithrombotic coating, creating a physical barrier that hinders protein and platelet adhesion that leads to heparin-binding thrombus

Subsystems:

?: phage loading / immobilization layer - ?

?: release or contact-kill mechanism - ?

?: surface substrate (coating architecture) - ?

?: stability preservation element - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: apply a marine-inspired anti-biofouling micro-textured surface (e.g., shark-skin riblet pattern) to the catheter lumen in conjunction with the existing antithrombotic coating, creating a physical barrier that hinders protein and platelet adhesion that leads to heparin-binding thrombus

Key Physics: Microbiology / phage anti-biofilm governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Microbiology / phage anti-biofilm governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

BIO-002: Exponential population dynamics. Expression: $N = N_0 * \exp(r * t)$. Applicability to this invention: CONDITIONAL; holds under: unlimited substrate; constant net rate over the window. Assumptions: kill kinetics on an immobilized surface are not well-mixed growth — re-measure before quantitative use. Engaged variables: NOT ESTABLISHED.

BIO-003: Surface-limited adsorption kinetics. Expression: $dN_{\text{infected}}/dt = k_{\text{ads}} * N_{\text{phage}} * N_{\text{bacteria}}$. Applicability to this invention: CONDITIONAL; holds under: well-mixed suspension; constant k_{ads} . Assumptions: surface-immobilized phage dosing is not well-mixed — use contact-area density instead (MODEL_DERIVED adaptation); k_{ads} is strain- and environment-specific. Engaged variables: NOT ESTABLISHED.

BIO-001: Multiplicity of infection. Expression: $MOI = N_{\text{phage}} / N_{\text{bacteria}}$. Applicability to this invention: CONDITIONAL; holds under: well-mixed system; planktonic or defined-surface assay. Assumptions: surface-immobilized phage dosing is not a well-mixed MOI — use contact-area density instead (MODEL_DERIVED adaptation). Engaged variables: NOT ESTABLISHED.

Assumptions:

- kill kinetics on an immobilized surface are not well-mixed growth — re-measure before quantitative use
- surface-immobilized phage dosing is not well-mixed — use contact-area density instead (MODEL_DERIVED adaptation)
- k_{ads} is strain- and environment-specific
- surface-immobilized phage dosing is not a well-mixed MOI — use contact-area density instead (MODEL_DERIVED adaptation)
- BIO-002 (Exponential population dynamics) is applicable only under: unlimited substrate; constant net rate over the window.
- BIO-003 (Surface-limited adsorption kinetics) is applicable only under: well-mixed suspension; constant k_{ads} .
- BIO-001 (Multiplicity of infection) is applicable only under: well-mixed system; planktonic or defined-surface assay.

- Model input variables requiring sourced values before any numeric result: N_0 (-), r (1/s), t (s), k_{ads} (m^3/s), N_{phage} (-), $N_{bacteria}$ (-), N_{phage} (-), $N_{bacteria}$ (-) — all UNKNOWN (no sourced value exists; Art. VIII).
- Model output variables produced by the governing model: N (-), $dN_{infected}/dt$ (1/s), MOI (-) — UNKNOWN until the inputs are sourced and the model is evaluated.
- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): surface phage loading density [N_{phage} , -] value_status=UNKNOWN; infectivity retention half-life [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; host range breadth [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; moi at contact surface [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; CFU challenge load [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; resistance emergence rate [r , 1/s] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'heparin-binding thrombus formation', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'infectivity loss before clinical use window', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'resistant strain emergence negating kill', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'host-range mismatch to clinical isolate', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'coating wear releasing loaded phage', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'biofilm matrix blocking phage access', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	solution must retain antithrombotic coating efficacy and biocompatibility	SOURCE_FACT
DI-002	Mechanism driver (proposed)	biofilm formation	MODELLED
DI-003	Expected effect target	the micro■ textured anti■ biofouling layer will reduce protein/platelet adhesion ■ diminish heparin■ binding thrombo	MODELLED
DI-004	Domain design input pattern: target organ applicability ■ FIVE FACTORS [biofilm]	never inferri	ENGINEERING_PROPOSED
DI-005	Domain design input pattern: device surface applicability ■ FIVE FACTORS [surface]	never inferri	ENGINEERING_PROPOSED
DI-006	Domain design input pattern: stability ■ EQUIVOCALITY ■ NOT SUPPLY ABANDONED (candidate pattern; confirm again)	never inferri	ENGINEERING_PROPOSED
DI-007	Regulatory design input: compliance ■ candidate standards ■ ISO 22196 (antimicrobial surface activity) ■ NO WG (PERFECT PROPOSED)	never inferri	ENGINEERING_PROPOSED
DI-008	Custodied observation: Pathology of catheters ■ https://relapex.com/patents/36268762 ■ catheter-associated infections ■ 105437e1f723e0f13a161	never inferri	ENGINEERING_PROPOSED
DI-009	Custodied observation: Hallmarks for Catheter-Associated Hemorrhagic Infection ■ https://pubs.bicomp-hem.org/article/6382011390-catheter-associated-hemorrhagic-infection ■ Disease Patterns ■ 682f2a567407ab49b0b90e67e0b1ec336	never inferri	ENGINEERING_PROPOSED
DI-010	Custodied observation: Over-the-Wire Release of Percutaneous Catheters ■ https://pubs.bicomp-hem.org/article/6382011390-catheter-associated-hemorrhagic-infection ■ Release ■ 74e24623285dd842a530e09fb06608fe45089730be3	never inferri	ENGINEERING_PROPOSED
DI-011	Custodied observation: Comprehensive Management of Hemodialysis Catheters ■ https://pubs.bicomp-hem.org/article/6382011390-catheter-associated-hemorrhagic-infection ■ Catheters ■ 1e87004e2185182040c06e34d3e35feaf9b58e66b8c949ef39	never inferri	ENGINEERING_PROPOSED

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	phage loading / immobilization layer realizing apply a marine■ inspired anti■ biofouling layer ■ textured surface (e.g., shark■ skin riblet pattern) to the catheter lumen	CONCEPTUAL	details of textured surface
DO-002	release or contact-kill mechanism realizing apply a marine■ inspired anti■ biofouling layer ■ textured surface (e.g., shark■ skin riblet pattern) to the catheter lumen	CONCEPTUAL	details of textured surface
DO-003	surface substrate (coating architecture) realizing apply a marine■ inspired anti■ biofouling layer ■ textured surface (e.g., shark■ skin riblet pattern) to the catheter lumen	CONCEPTUAL	details of textured surface
DO-004	stability preservation element realizing apply a marine■ inspired anti■ biofouling layer ■ textured surface (e.g., shark■ skin riblet pattern) to the catheter lumen	CONCEPTUAL	details of textured surface
DO-005	phage loading / immobilization layer interfaces with release or contact-kill mechanism ■ interface ■ signaling ■ information (minimal) ■ for applying a marine■ inspired anti■ biofouling layer	CONCEPTUAL	interface ■ signaling ■ information (minimal) ■ for applying a marine■ inspired anti■ biofouling layer
DO-006	release or contact-kill mechanism interfaces with surface ■ coating architecture ■ interface ■ signaling ■ information (minimal) ■ for applying a marine■ inspired anti■ biofouling layer	CONCEPTUAL	interface ■ signaling ■ information (minimal) ■ for applying a marine■ inspired anti■ biofouling layer
DO-007	surface substrate (coating architecture) interfaces with stability preservation element ■ interface ■ signaling ■ information (minimal) ■ for applying a marine■ inspired anti■ biofouling layer	CONCEPTUAL	interface ■ signaling ■ information (minimal) ■ for applying a marine■ inspired anti■ biofouling layer
DO-008	Device boundaries: patient/fluid inlet and outlet terminal ■ CONCEPTUAL control/data processing ■ required by applying a marine■ inspired anti■ biofouling micro■ textured surface	CONCEPTUAL	control/data processing ■ required by applying a marine■ inspired anti■ biofouling micro■ textured surface
DO-009	Flow/force/signal path geometry realizing the mechanism ■ information ■ minimal ■ dimensions ■ dimensions ■ external ■ surface (e.g., shark■ skin riblet pattern)	CONCEPTUAL	information ■ minimal ■ dimensions ■ dimensions ■ external ■ surface (e.g., shark■ skin riblet pattern)
DO-010	Operating range for surface phage loading density within which the mechanism ■ information ■ minimal ■ dimensions ■ dimensions ■ external ■ surface (e.g., shark■ skin riblet pattern)	UNKNOWN	information ■ minimal ■ dimensions ■ dimensions ■ external ■ surface (e.g., shark■ skin riblet pattern)
DO-011	Operating range for infectivity retention half-life within which the mechanism ■ information ■ minimal ■ dimensions ■ dimensions ■ external ■ surface (e.g., shark■ skin riblet pattern)	UNKNOWN	information ■ minimal ■ dimensions ■ dimensions ■ external ■ surface (e.g., shark■ skin riblet pattern)
DO-012	Operating range for host range breadth within which the mechanism ■ information ■ minimal ■ dimensions ■ dimensions ■ external ■ surface (e.g., shark■ skin riblet pattern)	UNKNOWN	information ■ minimal ■ dimensions ■ dimensions ■ external ■ surface (e.g., shark■ skin riblet pattern)
DO-013	Operating range for moi at contact surface within which the mechanism ■ information ■ minimal ■ dimensions ■ dimensions ■ external ■ surface (e.g., shark■ skin riblet pattern)	UNKNOWN	information ■ minimal ■ dimensions ■ dimensions ■ external ■ surface (e.g., shark■ skin riblet pattern)
DO-014	Operating range for CFU challenge load within which the mechanism ■ information ■ minimal ■ dimensions ■ dimensions ■ external ■ surface (e.g., shark■ skin riblet pattern)	UNKNOWN	information ■ minimal ■ dimensions ■ dimensions ■ external ■ surface (e.g., shark■ skin riblet pattern)
DO-015	Operating range for resistance emergence rate within which the mechanism ■ information ■ minimal ■ dimensions ■ dimensions ■ external ■ surface (e.g., shark■ skin riblet pattern)	UNKNOWN	information ■ minimal ■ dimensions ■ dimensions ■ external ■ surface (e.g., shark■ skin riblet pattern)
DO-016	Sensing -> decision -> actuation loop maintaining the mechanism ■ information ■ minimal ■ dimensions ■ dimensions ■ external ■ surface (e.g., shark■ skin riblet pattern)	PROPOSED	information ■ minimal ■ dimensions ■ dimensions ■ external ■ surface (e.g., shark■ skin riblet pattern)
DO-017	Information path: sensing element -> decision/monitor ■ CONCEPTUAL ■ act (minimal); rich data paths are NOT APPLICABLE to this mechanical/physical in	CONCEPTUAL	act (minimal); rich data paths are NOT APPLICABLE to this mechanical/physical in
DO-018	Fixture enabling: phage infectivity retention vs time on surface ■ CONCEPT ■ named, fixture geometry ■ UNKNOWN ■ selection; acceptance instrumentation	UNKNOWN	CONCEPT ■ named, fixture geometry ■ UNKNOWN ■ selection; acceptance instrumentation
DO-019	Fixture enabling: biofilm prevention CFU challenge assay ■ CONCEPT ■ named, fixture geometry ■ UNKNOWN ■ selection; acceptance instrumentation	UNKNOWN	CONCEPT ■ named, fixture geometry ■ UNKNOWN ■ selection; acceptance instrumentation
DO-020	Fixture enabling: host-range and resistance-emergence assay ■ CONCEPT ■ named, fixture geometry ■ UNKNOWN ■ selection; acceptance instrumentation	UNKNOWN	CONCEPT ■ named, fixture geometry ■ UNKNOWN ■ selection; acceptance instrumentation
DO-021	Fixture enabling: CFU biofilm-prevention challenge assay ■ CONCEPT ■ named, fixture geometry ■ UNKNOWN ■ selection; acceptance instrumentation	UNKNOWN	CONCEPT ■ named, fixture geometry ■ UNKNOWN ■ selection; acceptance instrumentation

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
surface phage loading density	UNKNOWN (no sourced value)	-	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
infectivity retention half-life	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
host range breadth	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
moi at contact surface	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
CFU challenge load	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
resistance emergence rate	UNKNOWN (no sourced value)	1/s	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
heparin-binding thrombosis	Failure of heparin-binding thrombosis	design information on heparin-binding thrombosis	On the basis of the design information, the design information is used to verify the design information.	UNKNOWN (no sourced value) and verification method
infectivity loss before release	Failure of infectivity loss before release	retention of infectivity loss before release	retention of infectivity loss before release	UNKNOWN (no sourced value) and verification method
resistant strain emergence	Failure of resistant strain emergence	heparin-binding thrombosis	heparin-binding thrombosis	UNKNOWN (no sourced value) and verification method
host-range mismatch	Failure of host-range mismatch	host-range mismatch	host-range mismatch	UNKNOWN (no sourced value) and verification method
coating wear releasing phage	Failure of coating wear releasing phage	coating wear releasing phage	coating wear releasing phage	UNKNOWN (no sourced value) and verification method
biofilm matrix blocking phage access	Failure of biofilm matrix blocking phage access	biofilm matrix blocking phage access	biofilm matrix blocking phage access	UNKNOWN (no sourced value) and verification method

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
heparin-binding thrombus formation	Failure of heparin-binding thrombus formation	design the device to be stable	Not Established
infectivity loss before clinical use	Failure of infectivity loss before clinical use	stabilization of phage particles	High infectivity loss before clinical use
resistant strain emergence	Failure of resistant strain emergence	negative phage selection	Not Established
host-range mismatch to clinical use	Failure of host-range mismatch to clinical use	isolation of phage's specificity	Not Established
coating wear releasing loaded phage	Failure of coating wear releasing loaded phage	phage-resistant coating	Not Established
biofilm matrix blocking phage access	Failure of biofilm matrix blocking phage access	degradation of phage access	Not Established

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	perform an in vitro blood flow loop comparison	in vitro blood flow loop comparison	compliance with ISO 10993-5	Not Established
VF-002	run killer experiment: falsification_test	run killer experiment: falsification_test	compliance with ISO 10993-5	Not Established
VF-003	phage infectivity retention vs time on surface	phage infectivity retention vs time on surface	compliance with ISO 10993-5	Not Established
VF-004	biofilm prevention CFU challenge assay	biofilm prevention CFU challenge assay	compliance with ISO 10993-5	Not Established
VF-005	CFU challenge assay against established biofilm	CFU challenge assay against established biofilm	compliance with ISO 10993-5	Not Established

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification)	compliance with ISO 10993-5	Not Established

11. Materials

Component	Candidate Material	Source	Verification Required	Status
domain candidate material	electrospun + metal-coated catheter	semi-proprietary device coatings	True	ENGINEERING_PROPOSED

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
electrospun + metal-coated catheter	domain candidate material	1	Material	electrospun + metal-coated catheter	semi-proprietary device coatings	High	Stability verification required

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- electrospinning + immobilization dip under aseptic control: domain registry manufacturing route - candidate only; no process qualification exists
- lyophilization/storage stabilization: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-001', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-002', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-003', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-004', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'value of critical parameter surface phage loading density [N_phage -]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter infectivity retention half-life [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter host range breadth [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter moi at contact surface [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter CFU challenge load [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}

- {'unknown': 'value of critical parameter resistance emergence rate [r 1/s]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}
- {'unknown': 'acceptance criterion for verification VF-001', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-001'}
- {'unknown': 'acceptance criterion for verification VF-002', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-002'}
- {'unknown': 'acceptance criterion for verification VF-004', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-004'}
- {'unknown': 'acceptance criterion for verification VF-005', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-005'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article	maintaining the proposed physical verification (to be required) status	perform an in vitro bioassay	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-02	constraint-compliance envelope article	per method	perform an in vitro bioassay	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-03	article for VF-001	per method	perform an in vitro bioassay	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-04	article for VF-002	per method	run killer experiment: NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-05	article for VF-003	per method	phage infectivity retention assay: NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-06	article for VF-004	per method	biofilm prevention CFU challenge assay: NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-07	article for VF-005	per method	CFU challenge assay: NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 7-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.